Executive Summary: Exos Aerospace Integrated Test Plan

SSLC Type-V Tank Supply & Sales-Commission Partnership and NASA RDRE Integration

---

Overview

Exos Aerospace is establishing a unified roadmap toward test-only orbital transfer vehicle (OTV) capability. The platform combines Exos's reusable suborbital rocket fleet and Integrated Pressurized Structures (IPS) — built on Type-V composite cryogenic tanks that Exos buys from its supply partner Scorpius Space Launch Company (SSLC) — with a ~5,800 lbf-class Rotating Detonation Rocket Engine (RDRE), down-selected via a NASA MSFC license vs. GTL machined-method trade study. Scorpius Space Launch Company (SSLC) is Exos's Type-V cryogenic tank supply partner and sales channel partner. Exos buys Type-V tanks from SSLC and integrates them into its Integrated Pressurized Structures and vehicles. Exos is also a sales channel for SSLC tank hardware — especially internationally and through the Exos Consortium — and earns a 15% sales commission on SSLC tank hardware it sells. Customers are quoted the same price SSLC quotes directly (no Exos markup). The commission also applies to SSLC sales to Exos, which effectively makes it a 15% rebate on the tanks Exos buys. Exos holds an 18% passive equity stake in SSLC. The commission terms were agreed with SSLC in principle and are being documented in a Sales Commission Agreement now in drafting.

The program introduces the BLK4 RDRE Test-Only OTV, deployed via rideshare missions on SpaceX Falcon 9 (Transporter), Rocket Lab Electron, or Firefly Alpha. Exos's suborbital rockets will serve as testbeds for RDRE-powered orbital components—reducing risk, validating subsystems in flight, and bridging to orbital operations without requiring new launch vehicle development.

Strategic Objectives

- **Accelerate RDRE Flight Validation: Mature the down-selected methane/oxygen RDRE (NASA MSFC license or GTL machined method) to TRL 8 by 2029 through suborbital and orbital demonstration flights.**

- **Leverage Exos Suborbital Fleet: Use Exos's reusable rockets to flight-test orbital-class propulsion modules, avionics, and recovery systems prior to barge or rideshare deployment.**
- **Integrate SSLC Type-V Tanks: Integrate SSLC-supplied Type-V tanks into Exos Integrated Pressurized Structures for 20–30% mass reduction and reusable OTV architectures, and sell SSLC tank hardware as SSLC's channel (especially international and Consortium customers).**
- **Utilize Texas-Based Infrastructure: Combine Midland, TX ground testing, Gulf of Mexico barge launches, and Exos's existing suborbital flight capabilities for a comprehensive, iterative development loop.**

Phased Implementation Plan

Funding close: Q4 2026 — enables Phase 1 start in January 2027.

Phase 1 (Q1 2027 – Q2 2027): Licensing & Early Integration

- **Run the RDRE trade study and down-select — NASA MSFC license (TechPort ID: 116281) vs. GTL machined RDRE method (Exos does not own the GTL technology; both are candidate paths); adapt combustion and injector designs for Exos integration.**
- **Integrate RDRE with SSLC-supplied Type-V pressurized tankage for the RDREW BLK3 test stage.**
- **Conduct early hot-fires and composite validation at Midland, TX facilities.**
- **Prepare Exos suborbital vehicle interfaces for in-flight RDRE component testing.**
- ***Milestones: License executed, PDR complete, first hot-fire by Q2 2027.***

Phase 2 (Q3 2027 – Q1 2028): Suborbital RDREW BLK3 Demonstrations

- **Conduct 5–7 suborbital flights from The Spaceport Company (TSC) Gulf barge to validate RDRE ignition, throttling, restart, and structural dynamics.**

- **Integrate RDRE propulsion modules and OTV avionics onto Exos suborbital rockets for partial-duration flight testing and recovery.**
- **Perform ground-based pre/post-flight analysis in Midland for composite evaluation.**
- ***Milestones:* 500+ seconds cumulative in-flight RDRE operation; TRL 7 achieved by end of 2027.**

Phase 3 (2028): BLK4 OTV Development and Rideshare Qualification (funded-future — proceeds only with program funding)

- **Develop BLK4 test-only OTV (100–500 kg) using RDRE propulsion and SSLC-supplied Type-V tanks integrated by Exos.**
- **Continue integration testing via Exos suborbital flights for propellant feed, guidance, and recovery systems prior to orbital rideshare readiness.**
- **Conduct extended-duration RDRE ground burns and vibration testing at Midland.**
- ***Milestones:* BLK4 CDR complete; first full-duration 5,800 lbf RDRE test by Q3 2028.**

Phase 4 (2028–2031): Orbital Demonstrations & Market Transition (funded-future)

- **Execute 3–5 rideshare launches (SpaceX, Rocket Lab, Firefly) to demonstrate RDRE-powered orbital maneuvers—orbit-raising, deorbit, and transfer simulations.**
- **Use Exos suborbital rockets between orbital missions for iterative design updates and subsystem validation.**
- **Position test results to support commercial OTV services for NASA, DoD, and commercial clients.**
- ***Milestones:* First orbital test Q1 2029; first OTV service contracts by Q3 2029.**

Financial Overview

Total Estimated Investment: $65–110 M over 4 years

- **NASA RDRE licensing: $1–5 M**
- **Gulf barge campaign: $5–10 M**
- **Rideshare missions: $15–25 M (3–5 deployments)**
- **Ground and suborbital testing (Midland ✦ Exos ops): $10–15 M**
- **Potential offsets via NASA/DoD programs (SBIR, CLPS, Tipping Point)**
- **SSLC Sales Commission (15%) — management-prepared base case: commission on Exos-originated third-party SSLC tank sales FY27 $0 · FY28 $37.5K · FY29 $75K · FY30 $113K · FY31 $150K; effective rebate on SSLC tanks Exos buys (credited against cost of sales) FY27 $68.1K · FY28 $90K · FY29 $135K · FY30 $180K · FY31 $225K; total commission economics FY27 $68.1K · FY28 $128K · FY29 $210K · FY30 $293K · FY31 $375K. No SSLC acquisition and no consolidated SSLC revenue are assumed; the 18% SSLC stake is carried passively at cost.**
- **Company projections (commission-only model, management-prepared): revenue FY27 $3.9M · FY28 $9.2M · FY29 $24.2M · FY30 $51.6M · FY31 $103.6M; EBITDA FY27 -$561K · FY28 -$1.1M · FY29 $4.7M · FY30 $16.1M · FY31 $55.9M. Fiscal years run July–June (FY27 = Jul 2026–Jun 2027). Management-prepared projections; not audited.**

Current Financial Position — FY2025 Actuals (per filed IRS Form 1120):

Revenue $1.49M; EBITDA ~$296K (~20% margin) — first profitable year (pre-tax income $189K).

Total assets $16.1M; ~$6.2M net operating loss carryforward (tax shield on future income).

Target Metrics:

- **90% RDRE efficiency demonstrated**
- **80% hardware recovery rate**
- **✦150 kg pressurized structure dry mass**
- **3✦ successful rideshare tests by 2029**
- **Commercial OTV-ready platform by 2031**

Testing and Trade Study Summary

A comparative analysis of TSC Gulf barge operations and Midland, TX test facilities confirms a hybrid strategy:

- **Barge: Optimal for flight safety, recovery operations, and FAA-approved offshore campaigns.**
- **Midland: Best for ground integration, engine qualification, and suborbital-to-orbital test transitions.**
- **Exos Suborbital Fleet: Critical bridge for real-flight testing of RDRE components and orbital vehicle systems prior to full orbital deployments.**
  → Recommendation: Midland for ground and subsystem integration; Exos suborbital flights for validation; TSC barge for full-system dynamic testing.

Strategic Advantages

- **Integrated Flight Test Path: Ground → Suborbital → Barge → Rideshare, minimizing risk and cost per iteration.**
- **RDRE Advantage: methane/oxygen detonation engine (~ 350 s Isp potential) via the down-selected path — NASA MSFC license or GTL machined method.**
- **Reusable Infrastructure: Exos suborbital platforms enable rapid reuse and real-world data collection.**
- **Cost Efficiency: Rideshare strategy reduces orbital access costs by ~ 60% versus dedicated launch.**
- **Texas Ecosystem Synergy: Co-located testing, composites manufacturing, and workforce availability.**

Next Actions (Q4 2026)

- **Execute the SSLC Sales Commission Agreement (15% on SSLC tank hardware sold through Exos at SSLC's quoted price, including SSLC sales to Exos); run**

the RDRE trade study and down-select (NASA MSFC license vs. GTL machined method).

- **Modify Exos suborbital vehicles for integrated RDRE subsystem testing.**
- **Secure TSC barge access and rideshare LOIs (SpaceX, Rocket Lab, Firefly).**
- **Begin composite structure and avionics PDR integration at Midland.**